

10. (a) We must have

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 \frac{c}{(1-x)^{1/2}} dx = 2c = 1.$$

Therefore $c = 1/2$. This p.d.f. has the appearance of Fig. S.3.5.

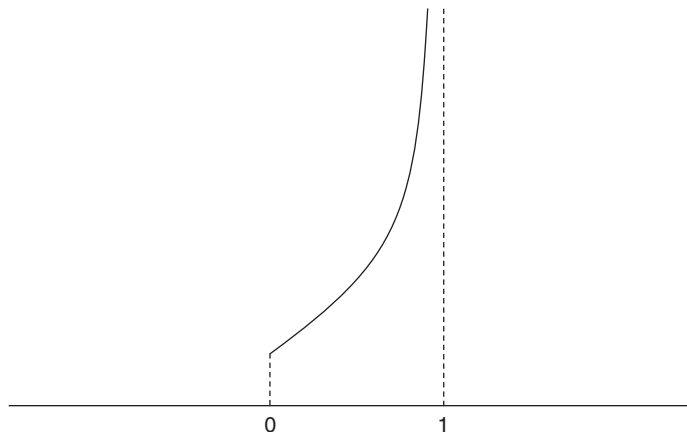


Figure S.3.5: Figure for Exercise 10a of Sec. 3.2.

It should be noted that although the values of $f(x)$ become arbitrarily large in the neighborhood of $x = 1$, the total area under the curve is equal to 1. It is seen, therefore, that the values of a p.d.f. can be greater than 1 and, in fact, can be arbitrarily large.

(b) $\int_0^{1/2} f(x) dx = 1 - (1/2)^{1/2}.$

11. Since $\int_0^1 (1/x) dx = \infty$, there is no constant c such that $\int_0^1 f(x) dx = 1$.

12. We shall find the c.d.f. of Y and evaluate it at 50. The c.d.f. of a random variable Y is $F(y) = \Pr(Y \leq y)$. In Fig. 3.1, on page 94 of the text, the event $\{Y \leq y\}$ has area $(y - 1) \times (200 - 4) = 196(y - 1)$ if $1 \leq y \leq 150$. We need to divide this by the area of the entire rectangle, 29,204. The c.d.f. of Y is then

$$F(y) = \begin{cases} 0 & \text{for } y < 1, \\ \frac{196(y - 1)}{29204} & \text{for } 1 \leq y \leq 150, \\ 1 & \text{for } y > 150. \end{cases}$$

So, in particular, $\Pr(Y \leq 50) = 0.3289$.

13. We find $\Pr(X < 20) = \int_0^{20} cxdx = 200c$. Setting this equal to 0.9 yields $c = 0.0045$.

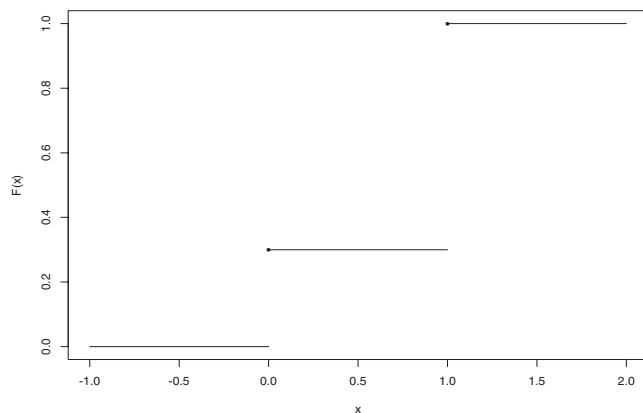
3.3 The Cumulative Distribution Function

Commentary

This section includes a discussion of quantile functions. These arise repeatedly in the construction of hypothesis tests and confidence intervals later in the book.

Solutions to Exercises

1. The c.d.f. $F(x)$ of X is 0 for $x < 0$. It jumps to $0.3 = \Pr(X = 0)$ at $x = 0$, and it jumps to 1 and stays there at $x = 1$. The c.d.f. is sketched in Fig. S.3.6.

Figure S.3.6: C.d.f. of X in Exercise 1 of Sec. 3.3.

2. The c.d.f. must have the appearance of Fig. S.3.7.

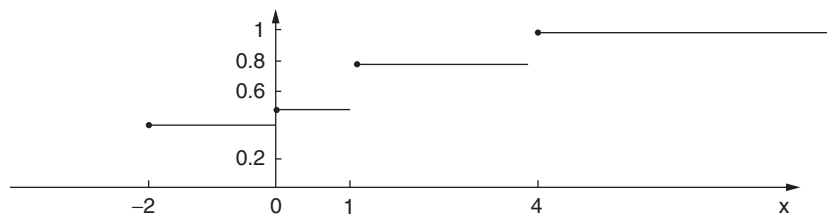


Figure S.3.7: C.d.f. for Exercise 2 of Sec. 3.3.

3. Here $\Pr(X = n) = 1/2^n$ for $n = 1, 2, \dots$. Therefore, the c.d.f. must have the appearance of Fig. S.3.8.

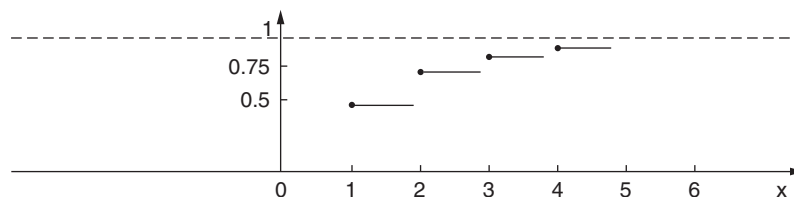


Figure S.3.8: C.d.f. for Exercise 3 of Sec. 3.3.

4. The numbers can be read off of the figure or found by subtracting two numbers off of the figure.
- The jump at $x = -1$ is $F(-1) - F(-1^-) = 0.1$.
 - The c.d.f. to the left of $x = 0$ is $F(0^-) = 0.1$.
 - The c.d.f. at $x = 0$ is $F(0) = 0.2$.

(d) There is no jump at $x = 1$, so $\Pr(X = 1) = 0$.

(e) $F(3) - F(0) = 0.6$.

(f) $F(3^-) - F(0) = 0.4$.

(g) $F(3) - F(0^-) = 0.7$.

(h) $F(2) - F(1) = 0$.

(i) $F(2) - F(1^-) = 0$.

(j) $1 - F(5) = 0$.

(k) $1 - F(5^-) = 0$.

(l) $F(4) - F(3^-) = 0.2$.

$$5. \quad f(x) = \frac{dF(x)}{dx} = \begin{cases} 0 & \text{for } x < 0, \\ \frac{2}{9}x & \text{for } 0 < x < 3, \\ 0 & \text{for } x > 3. \end{cases}$$

The value of $f(x)$ at $x = 0$ and $x = 3$ is irrelevant. This p.d.f. has the appearance of Fig. S.3.9.

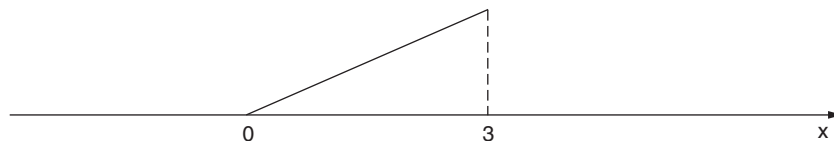


Figure S.3.9: Figure for Exercise 5 of Sec. 3.3.

$$6. \quad f(x) = \frac{dF(x)}{dx} = \begin{cases} \exp(x - 3) & \text{for } x < 3, \\ 0 & \text{for } x > 3. \end{cases}$$

The value of $f(x)$ at $x = 3$ is irrelevant. This p.d.f. has the appearance of Fig. S.3.10.

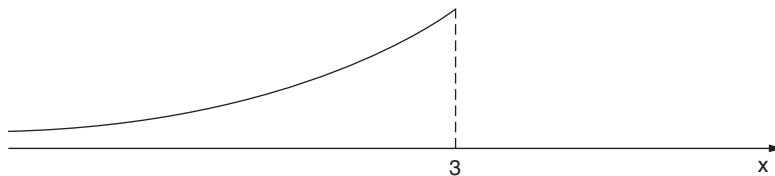


Figure S.3.10: Figure for Exercise 6 of Sec. 3.3.

It should be noted that although this p.d.f. is positive over the unbounded interval where $x < 3$, the total area under the curve is finite and is equal to 1.

7. The c.d.f. equals 0 for $x < -2$ and it equals 1 for $x > 8$. For $-2 \leq x \leq 8$, the c.d.f. equals

$$F(x) = \int_{-2}^x \frac{dy}{10} = \frac{x + 2}{10}.$$

The c.d.f. has the appearance of Fig. S.3.11.

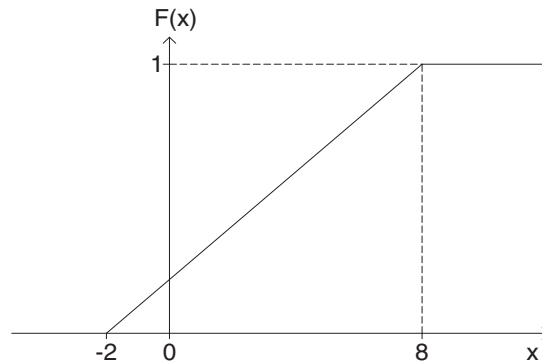


Figure S.3.11: Figure for Exercise 7 of Sec. 3.3.

8. $\Pr(Z \leq z)$ is the probability that Z lies within a circle of radius z centered at the origin. This probability is

$$\frac{\text{Area of circle of radius } z}{\text{Area of circle of radius } 1} = z^2, \quad \text{for } 0 \leq z \leq 1.$$

The c.d.f. is plotted in Fig. S.3.12.

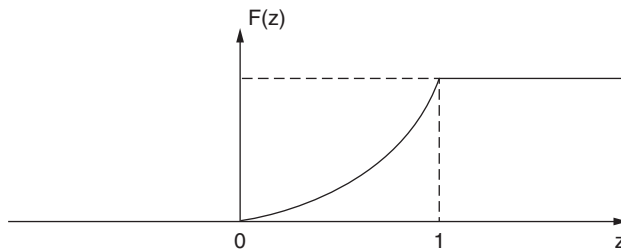


Figure S.3.12: C.d.f. for Exercise 8 of Sec. 3.3.

9. $\Pr(Y = 0) = \Pr(X \leq 1) = 1/5$ and $\Pr(Y = 5) = \Pr(X \geq 3) = 2/5$. Also, Y is distributed uniformly between $Y = 1$ and $Y = 3$, with a total probability of $2/5$. Therefore, over this interval $F(y)$ will be linear with a total increase of $2/5$. The c.d.f. is plotted in Fig. S.3.13.
10. To find the quantile function $F^{-1}(p)$ when we know the c.d.f., we can set $F(x) = p$ and solve for x .

$$\frac{x}{1+x} = p; \quad x = p + px; \quad x(1-p) = p; \quad x = \frac{p}{1-p}.$$

The quantile function is $F^{-1}(p) = p/(1-p)$.

11. As in Exercise 10, we set $F(x) = p$ and solve for x .

$$\frac{1}{9}x^2 = p; \quad x^2 = 9p; \quad x = 3p^{1/2}.$$

The quantile function of X is $F^{-1}(p) = 3p^{1/2}$.

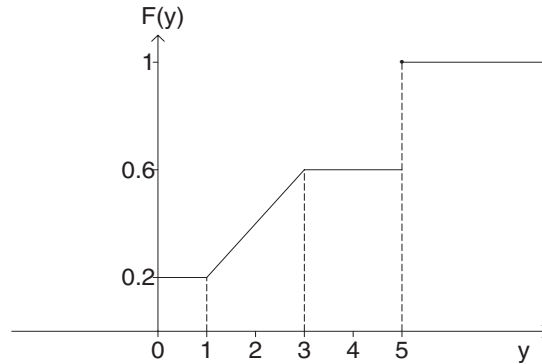


Figure S.3.13: C.d.f. for Exercise 9 of Sec. 3.3.

12. Once again, we set $F(x) = p$ and solve for x .

$$\exp(x - 3) = p; \quad x - 3 = \log(p); \quad x = 3 + \log(p).$$

The quantile function of X is $F^{-1}(p) = 3 + \log(p)$.

13. VaR at probability level 0.95 is the negative of the 0.05 quantile. Using the result from Example 3.3.8, the 0.05 quantile of the uniform distribution on the interval $[-12, 24]$ is $0.05 \times 24 - 0.95 \times 12 = -10.2$. So, VaR at probability level 0.95 is 10.2
14. Using the table of binomial probabilities in the back of the book, we can compute the c.d.f. F of the binomial distribution with parameters 10 and 0.2. We then find the first values of x such that $F(x) \geq 0.25$, $F(x) \geq 0.5$, and $F(x) \geq 0.75$. The first few distinct values of the c.d.f. are

x	0	1	2	3
$F(x)$	0.0174	0.3758	0.6778	0.8791

So, the quartiles are 1 and 3, while the median is 2.

15. Since $f(x) = 0$ for $x \leq 0$ and for $x \geq 1$, the c.d.f. $F(x)$ will be flat (0) for $x \leq 0$ and flat (1) for $x \geq 1$. Between 0 and 1, we compute $F(x)$ by integrating the p.d.f. For $0 < x < 1$,

$$F(x) = \int_0^x 2y dy = x^2.$$

The requested plot is identical to Fig. S.3.12 for Exercise 8 in this section.

16. For each $0 < p < 1$, we solve for x in the equation $F(x) = p$, with F specified in (3.3.2):

$$\begin{aligned} p &= 1 - \frac{1}{1+x}, \\ \frac{1}{1-p} &= 1+x, \\ \frac{1}{1-p} - 1 &= x. \end{aligned}$$

The quantile function is $F^{-1}(p) = 1/(1-p) - 1$ for $0 < p < 1$.

17. (a) Let $0 < p_1 < p_2 < 1$. Define $A_i = \{x : F(x) \geq p_i\}$ for $i = 1, 2$. Since $p_1 < p_2$ and F is nondecreasing, it follows that $A_2 \subset A_1$. Hence, the smallest number in A_1 (which equals $F^{-1}(p_1)$ by definition) is no greater than the smallest number in A_2 (which equals $F^{-1}(p_2)$ by definition). That is, $F^{-1}(p_1) \leq F^{-1}(p_2)$, and the quantile function is nondecreasing.
- (b) Let $x_0 = \lim_{\substack{p \rightarrow 0 \\ p > 0}} F^{-1}(p)$. We are asked to prove that x_0 is the greatest lower bound of the set $C = \{c : F(c) > 0\}$. First, we show that no $x > x_0$ is a lower bound on C . Let $x > x_0$ and $x_1 = (x + x_0)/2$. Then $x_0 < x_1 < x$. Because $F^{-1}(p)$ is nondecreasing, it follows that there exists $p > 0$ such that $F^{-1}(p) < x_1$, which in turn implies that $p \leq F(x_1)$, and $F(x_1) > 0$. Hence $x_1 \in C$, and x is not a lower bound on C . Next, we prove that x_0 is a lower bound on C . Let $x \in C$. We need only prove that $x_0 \leq x$. Because $F^{-1}(p)$ is nondecreasing, we must have $\lim_{\substack{p \rightarrow 0 \\ p > 0}} F^{-1}(p) \leq F^{-1}(q)$ for all $q > 0$. Hence, $x_0 \leq F^{-1}(p)$ for all $q > 0$. Because $x \in C$, we have $F(x) > 0$. Let $q = F(x)$ so that $q > 0$. Then $x_0 \leq F^{-1}(q) \leq x$. The proof that x_1 is the least upper bound on the set of all d such that $F(d) < 1$ is very similar.
- (c) Let $0 < p < 1$. Because F^{-1} is nondecreasing, $F^{-1}(p^-)$ is the least upper bound on the set $C = \{F^{-1}(q) : q < p\}$. We need to show that $F^{-1}(p)$ is also that least upper bound. Clearly, $F^{-1}(p)$ is an upper bound, because F^{-1} is nondecreasing and $p > q$ for all $q < p$. To see that $F^{-1}(p)$ is the least upper bound, let y be an upper bound. We need to show $F^{-1}(p) \leq y$. By definition, $F^{-1}(p)$ is the greatest lower bound on the set $D = \{x : F(x) \geq p\}$. Because y is an upper bound on C , it follows that $F^{-1}(q) \leq y$ for all $q < p$. Hence, $F(y) \geq q$ for all $q < p$. Because F is nondecreasing, we have $F(y) \geq p$, hence $y \in D$, and $F^{-1}(p) \leq y$.
18. We know that $\Pr(X = c) = F(c) - F(c^-)$. We will prove that $p_1 = F * (c)$ and $p_0 = F(c^-)$. For each $p \in (0, 1)$ define
- $$C_p = \{x : F(x) \geq p\}.$$
- Condition (i) says that, for every $p \in (p_0, p_1)$, c is the greatest lower bound on the set C_p . Hence $F(c) \geq p$ for all $p < p_1$ and $F(c) \geq p_1$. If $F(c) > p_1$, then for $p = (p_1 + F(c))/2$, $F^{-1}(p) \leq c$, and condition (iii) rules this out. So $F(c) = p_1$. The rest of the proof is broken into two cases. First, if $p_0 = 0$, then for every $\epsilon > 0$, c is the greatest lower bound on the set C_ϵ . This means that $F(x) < \epsilon$ for all $x < c$. Since this is true for all $\epsilon > 0$, $F(x) = 0$ for all $x < c$, and $F(c^-) = 0 = p_0$. For the second case, assume $p_0 > 0$. Condition (ii) says $F^{-1}(p_0) < c$. Since $F^{-1}(p_0)$ is the greatest lower bound on the set C_{p_0} , we have $F(x) < p_0$ for all $x < c$. Hence, $p_0 \geq F(c^-)$. Also, for all $p < p_0$, $p \leq F(c^-)$, hence $p_0 \leq F(c^-)$. Together, the last two inequalities imply $p_0 = F(c^-)$.
19. First, we show that $F^{-1}(F(x)) \leq x$. By definition $F^{-1}(F(x))$ is the smallest y such that $F(y) \geq F(x)$. Clearly $F(x) \geq F(x)$, hence $F^{-1}(F(x)) \leq x$. Next, we show that, if $p > F(x)$, then $F^{-1}(p) > x$. Let $p > F(x)$. By Exercise 17, we know that $F^{-1}(p) \geq x$. By definition, $F^{-1}(p)$ is the greatest lower bound on the set $C_p = \{y : F(y) \geq p\}$. All $y \in C_p$ satisfy $F(y) > (p + F(x))/2$. Since F is continuous from the right, $F(F^{-1}(p)) \geq (p + F(x))/2$. But $F(x) < (p + F(x))/2$, so $x \neq F^{-1}(p)$, hence $F^{-1}(p) > x$.
20. Figure S.3.14 has the plotted c.d.f., which equals $0.0045x^2/2$ for $0 < x < 20$. On the plot, we see that $F(10) = 0.225$.

3.4 Bivariate Distributions

Commentary

The bivariate distribution function is mentioned at the end of this section. The only part of this discussion that is used later in the text is the fact that the joint p.d.f. is the second mixed partial derivative of the